

00-frontmatter

Helios — Project Bible

Autonomous Growth Diagnosis Engine

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Purpose. This document is the authoritative, rebuild-from-scratch reference for Helios — an autonomous "AI Growth Analyst" that continuously diagnoses *why* an e-commerce funnel is moving, distinguishes mix-shift from rate-change, quantifies revenue-at-risk in dollars, prescribes a prioritized and statistically-defensible experiment backlog, and produces an executive Decision Brief, all grounded in governed SQL it writes, verifies, and is graded on by an offline evaluation benchmark. It is written so that an engineer with no prior context could reconstruct the entire system — data model, semantic layer, MCP tool surface, seven-agent orchestration, memory, evaluation harness, and roadmap — years from now. Where any section disagrees with the Canonical Reference Card below, the Reference Card (which mirrors the project Foundation spec) wins.

How to Read This Document

The Bible is organized in 25 numbered sections grouped into nine thematic parts. Read it top-to-bottom for the full narrative, or jump by role:

- **New to Helios?** Read Part I (Strategy, §1–5) for the *why*, then the Canonical Reference Card in this front matter for the shared vocabulary. Those two together are enough to follow any later section.
- **Analytics / data engineer?** Focus on Part III (Data Model, §8–11), Part IV (Semantics, §12–14), and Part V (Analytics Engineering, §15–17). These define every metric, the dbt DAG, and the BigQuery cost controls.
- **AI / platform engineer?** Focus on Part VI (MCP, §18) and Part VII (Agents + Memory, §19 & §22). The five MCP servers are the trust boundary; the seven agents are the control plane.
- **Evaluating rigor / trust?** Read Part VIII (Evaluation + Experimentation, §20–21): the labeled benchmark, the 85%-vs-45% claim, and the experiment-design framework.

- **Hiring manager / interviewer?** Part IX (Roadmap + Narrative, §23–25) maps the maturity ladder and the resume/interview framing.

Three reading aids: (1) every term is defined once in the **Glossary** at the end of this front matter; (2) every canonical name (metric, dimension, MCP tool, agent) is enumerated in the **Canonical Reference Card** and must never be paraphrased elsewhere; (3) requirements carry stable IDs (FR-, NFR-) so they can be cited and traced.

Table of Contents

Sections are listed in thematic reading order. Note that **Memory Architecture (22)** is presented alongside **Agent Architecture (19)** in Part VII, because memory is the state plane that the agent control plane reads from and writes to — they are best understood together rather than in strict numeric order.

#	Section	Part
Part I — Strategy		
1	Product Vision	Strategy
2	Business Problem	Strategy
3	User Personas	Strategy
4	Core Product Thesis	Strategy
5	Success Metrics	Strategy
Part II — Requirements		
6	Functional Requirements	Requirements
7	Non-Functional Requirements	Requirements
Part III — Data Model		
8	Data Model	Data Model
9	Event Model	Data Model
10	Funnel Definitions	Data Model
11	Revenue Definitions	Data Model
Part IV — Semantics		

#	Section	Part
12	Channel Attribution Definitions	Semantics
13	Metric Definitions	Semantics
14	Semantic Layer Design	Semantics
Part V — Analytics Engineering		
15	Analytics Engineering Architecture	Analytics Engineering
16	dbt Project Structure	Analytics Engineering
17	BigQuery Architecture	Analytics Engineering
Part VI — MCP		
18	MCP Architecture	MCP
Part VII — Agents & Memory		
19	Agent Architecture	Agents
22	Memory Architecture (<i>presented alongside §19</i>)	Agents
Part VIII — Evaluation & Experimentation		
20	Evaluation Framework	Eval / Experiment
21	Experimentation Framework	Eval / Experiment
Part IX — Roadmap & Narrative		
23	Future Roadmap	Roadmap / Narrative
24	Interview Narrative	Roadmap / Narrative
25	Resume Narrative	Roadmap / Narrative

Document Conventions

Naming. All physical names — metrics, dimensions, dbt models, columns, MCP servers, MCP tools — are `snake_case` and must match the Canonical Reference Card exactly. Never invent a synonym (e.g. `conversion_pct` for `session_conversion_rate`); unknown names are a hard error at the semantic layer, not a fallback. MCP servers are named `<area>-mcp` (e.g. `warehouse-mcp`). dbt layers use fixed prefixes: `stg_<source>__<entity>` (staging), `int_<source>__<entity>` (intermediate), `fct_*` / `dim_*` (marts). Agent names are capitalized proper nouns (Orchestrator, Monitor, Decompose, Diagnose, Prescribe, Narrator, Critic).

Code fences. Fenced blocks are labeled by intent: `sql` for governed/dbt SQL, `python` for MCP-server or harness code, `yaml` for configuration and semantic-layer definitions, `json` for MCP request/response payloads, and `text` for diagrams, ASCII funnels, and pseudocode. SQL shown in the Bible is illustrative of the *governed* definition; at runtime the LLM never authors it — `semantic-mcp.build_query` emits it.

Requirement numbering. Functional requirements are `FR-<domain-letter><n>` (e.g. `FR-A1` in domain A "Data & Semantic Layer"). Non-functional requirements are `NFR-<area-letter><n>` (e.g. `NFR-P1` for Performance). Each carries a MoSCoW priority (Must / Should / Could / Won't-for-now) and a verification method. Findings, hypothesis cards, scenarios, and experiments carry their own ID prefixes (`F-*` , `H-*` , `S###` , `Exp-###`).

Conventions for money and rates. All money uses GA4's `_in_usd` columns; non-USD twins are never aggregated. Rates are computed as `SUM(numerator)/SUM(denominator)` after grouping (never an average of per-segment ratios) — this re-aggregation discipline is itself the defense against Simpson's paradox.

Canonical Reference Card

This card is the single source of truth for shared vocabulary. It mirrors the project Foundation spec exactly. Any section that disagrees with this card is in error.

Macro funnel stages

```
session_start → view_item → add_to_cart → begin_checkout → add_shipping_info → add_payment_info
```

Helios reports **step-to-step** conversion (each stage / prior stage) AND the **overall**

```
session_conversion_rate = purchasing_sessions / sessions
```

 . The funnel is session-scoped;

the session key is (user_pseudo_id, ga_session_id) .

#	Stage	Maps to event
0	session_start	session_start (denominator anchor)
1	view_item	view_item
2	add_to_cart	add_to_cart
3	begin_checkout	begin_checkout
4	add_shipping_info	add_shipping_info
5	add_payment_info	add_payment_info
6	purchase	purchase

Canonical metric names (snake_case)

Group	Metrics
Volume	sessions, users, new_users, returning_users, engaged_sessions, view_item_sessions, add_to_cart_sessions, begin_checkout_sessions, purchasing_sessions
Engagement	engagement_rate
Funnel rate	session_conversion_rate (= purchasing_sessions/sessions), view_to_cart_rate, cart_to_checkout_rate, checkout_to_purchase_rate, cart_abandonment_rate, checkout_abandonment_rate
Revenue	transactions (distinct transaction_id), revenue (sum purchase_revenue_in_usd), gross_revenue, net_revenue (gross – refunds), aov (= revenue/transactions), items_per_transaction
Efficiency	revenue_per_session (RPS = revenue/sessions), revenue_per_user (ARPU = revenue/users)

Canonical dimensions

device_category, operating_system, browser, country, region, channel_group, source, medium, campaign, landing_page (first page_location of session), item_category, item_name, is_new_user, day, week, session_number_bucket .

Channel groups (GA4-style default grouping): Direct, Organic Search, Paid Search, Display, Paid Social, Organic Social, Email, Affiliates, Referral, Other.

The five MCP servers and their tools

Server	Role	Tools
warehouse-mcp	Governed execution (only holder of a BigQuery client)	<code>list_tables</code> , <code>describe_table</code> , <code>dry_run(sql)</code> , <code>run_query(sql)</code> , <code>reconcile(metric, grain)</code>
semantic-mcp	The ONLY path to SQL (anti-hallucination layer)	<code>get_metric(name)</code> , <code>list_dimensions()</code> , <code>build_query(metric, dims, filters, window)</code>
stats-mcp	The ONLY path to math (determinism)	<code>detect_anomaly(series, method)</code> , <code>decompose_change(metric, dim, t0, t1)</code> , <code>significance_test(a, b)</code> , <code>forecast(series, horizon)</code> , <code>cohort_retention(...)</code> , <code>rfm_segment(...)</code>
experiment-mcp	Statistically-defensible backlog	<code>power_analysis(baseline, mde, alpha, power)</code> , <code>runtime_estimate(n, traffic)</code> , <code>design_experiment(hypothesis, metric)</code>
report-mcp	Narration and memory	<code>render_brief(findings)</code> , <code>save_diagnosis(...)</code> , <code>recall_prior(metric, segment)</code> , <code>export(format)</code>

The seven agents and their roles

Agent	Model	Role
Orchestrator (Planner)	Opus	Plan the run, sequence stages, manage budget, route to Critic, finalize
Monitor	Sonnet	Anomaly detection over canonical metric series
Decompose	Sonnet	Split each anomaly into mix-shift vs rate-change vs interaction
Diagnose	Opus	Hypothesis-tree root-cause analysis, each branch SQL-verified
Prescribe	Sonnet	Convert confirmed root causes into a prioritized, powered experiment backlog
Narrator	Sonnet	Compose the executive Decision Brief from surviving findings

Agent	Model	Role
Critic	Opus	Adversarially refute every finding (confound, sample, seasonality, data quality) before it ships

Core decomposition identity (the technical centerpiece). For an aggregate rate $R = \sum_i (w_i \cdot r_i)$ where w_i is segment volume share and r_i is segment rate, the change ΔR from $t_0 \rightarrow t_1$ decomposes as:

```

mix effect    =  $\sum_i ( \Delta w_i \cdot r_i(t_0) )$       # traffic composition changed
rate effect   =  $\sum_i ( w_i(t_0) \cdot \Delta r_i )$       # in-segment behavior changed
interaction   =  $\sum_i ( \Delta w_i \cdot \Delta r_i )$       # both moved together
 $\Delta R$       = mix effect + rate effect + interaction

```

Success-metric targets: root-cause accuracy $\geq 85\%$ (vs $\leq 45\%$ naive baseline); time-to-diagnosis < 5 min/run; 0 hallucinated columns/metrics (100% governed SQL); 100% of findings carry significance + dollar impact; query cost per run under a fixed BigQuery byte budget.

Maturity levels: L1 Intern MVP; L2 Strong portfolio project; L3 Top-1% undergraduate project.

Glossary

Term / acronym	Definition
Action tracking	A memory store recording whether a prescribed experiment shipped and whether the fix actually worked (status $\in$ proposed / shipped / running / completed / abandoned, plus observed lift and outcome). Closes the "did-the-fix-work" loop.
AOV	Average Order Value = $\text{revenue} / \text{transactions}$. Reflects merchandise value (excludes shipping and tax).
ARPU	Average Revenue Per User = $\text{revenue} / \text{users}$ (= revenue_per_user). Denominator is all users in the window, not just purchasers.
Anomaly detection	Deterministic flagging of abnormal movement in a metric series via <code>stats-mcp.detect_anomaly</code> (STL-residual, EWMA, or robust z-score), accounting for weekly seasonality.
Channel group	GA4-style default channel grouping (Direct, Organic Search, Paid Search, Display, Paid Social, Organic Social, Email, Affiliates, Referral, Other) derived from session-scoped source/medium.

Term / acronym	Definition
Cohort	A set of users grouped by the week of their <code>user_first_touch_timestamp</code> ; the basis for retention curves.
Critic	The adversarial verifier agent (Opus) that attempts to refute every finding before it ships; verdicts are PASS / DOWNGRADE / DROP.
Decision Brief	The primary product surface: the autonomously-produced, executive-ready artifact rendered by <code>report-mcp.render_brief</code> , where every finding carries a decomposition, a significance test, a dollar revenue-at-risk, and a recommended action.
Decomposition	Splitting an aggregate rate change ΔR into mix effect, rate effect, and interaction (see the core identity above); the technical centerpiece, computed by <code>stats-mcp.decompose_change</code> .
DiD (difference-in-differences)	A quasi-experimental readback: $DiD = (treated_post - treated_pre) - (control_post - control_pre)$, netting out seasonality via a comparable control segment, with parallel-trends checked on the pre-period.
dry-run	A BigQuery cost/schema check (<code>warehouse-mcp.dry_run</code>) that returns estimated scanned bytes without executing or billing; mandatory before any <code>run_query</code> to enforce the byte budget.
Engaged session	A GA4-defined engaged session (session-scoped <code>session_engaged = '1'</code> , with the GA4 fallback of engagement time / multiple page views / a conversion event); <code>engagement_rate = engaged_sessions / sessions</code> .
Faithfulness	Whether the Narrator's prose claims are entailed by the underlying SQL evidence and <code>stats-mcp</code> outputs (target ≥ 0.95); checked by rule + Critic-as-judge entailment.
First-touch attribution	Crediting a session/transaction to the channel of the user's first session ever; stored on <code>dim_users.first_touch_channel</code> .
<code>fct_daily_funnel</code>	The pre-aggregated daily funnel fact (additive step counts + revenue per day $\times$ dimension cell); the primary feed for Monitor and Decompose.
<code>ga_session_id</code>	A session id that is unique only <i>within</i> a <code>user_pseudo_id</code> (never globally); lives inside <code>event_params</code> and must be unnested. Half of the session key.
Governed SQL	SQL emitted exclusively by <code>semantic-mcp</code> from versioned metric/dimension definitions; the LLM never authors raw SQL. Guarantees 0 hallucinated columns.

Term / acronym	Definition
Grounding over generation	The first principle: the LLM composes governed metrics and calls deterministic tools, never authoring SQL or computing statistics in token-space.
Hallucination rate	The count of columns/metrics in emitted SQL not present in the semantic registry or GA4 schema; a hard-zero CI gate.
ICE / PIE	Experiment-prioritization scores. ICE = (Impact × Confidence) / Effort; PIE = Potential × Importance × Ease. Both are deterministic and recorded on each hypothesis card.
Idempotency	Re-running a model/run over the same partitions yields byte-identical output (via partition-by-date + <code>insert_overwrite</code>); no duplicate diagnoses per window.
Interaction effect	The third decomposition term, $\sum \Delta w_i \cdot \Delta r_i$ — change attributable to weight and rate moving together.
Last-touch attribution	Crediting to the channel of the converting session itself; the Helios default for session-grained funnel and revenue diagnosis.
Last-non-direct attribution	Crediting to the most recent non- (direct) channel within a lookback window (GA4 default 90 days).
MCP (Model Context Protocol)	An open client-server protocol over JSON-RPC that lets the agent runtime invoke schema-typed external tools; the mechanism that makes the grounding boundary physically enforceable.
MDE	Minimum Detectable Effect — the smallest effect (often relative, e.g. 10%) an experiment is powered to detect; an input to <code>power_analysis</code> .
Memory store	Durable state across runs: diagnosis history (+ embeddings), suppression list, business glossary, seasonality/launch calendars, action tracking, and the run-state/audit trail.
Mix-shift (mix effect)	A change in an aggregate rate caused by traffic <i>composition</i> shifting between segments (weights w_i), with in-segment behavior unchanged. The non-causal half that naive RCA mistakes for behavior change.
Quasi-experiment	An inference design used when no live A/B traffic exists (this dataset is observational): pre/post and difference-in-differences readbacks via <code>stats-mcp</code> .
Rate-change (rate effect)	A change in an aggregate rate caused by in-segment <i>behavior</i> shifting (segment rates r_i), with composition held fixed. The genuinely causal half.

Term / acronym	Definition
Reconciliation	Checking that a fact-derived metric matches <code>warehouse-mcp.reconcile</code> canonical totals within tolerance ($\leq 0.5\%$); part of verify-then-trust.
Revenue-at-risk	The dollar counterfactual every finding carries: dollars recoverable if a degraded rate returned to its t0 / forecast baseline ($\approx \Delta\text{rate} \times \text{affected sessions} \times \text{downstream value}$).
RFM	Recency / Frequency / Monetary segmentation, scored into quintiles per user by <code>stats-mcp.rfm_segment</code> .
RPS	Revenue Per Session = <code>revenue / sessions</code> (= <code>revenue_per_session</code>); decomposes exactly as <code>session_conversion_rate × aov</code> .
Sessionization	Reconstructing a session row from events sharing the same (<code>user_pseudo_id</code> , <code>ga_session_id</code>) — GA4 ships no session row.
Simpson's paradox	When an aggregate rate moves opposite to every segment's rate because the mix shifted; the failure mode the decomposition is built to detect.
Suppression list	A memory store of acknowledged/ignored causes that must not be re-raised, with a TTL so re-emerging issues eventually re-surface.
<code>traffic_source</code> gotcha	The event-level <code>traffic_source</code> struct is USER-LEVEL first-touch attribution, not session source; Helios prefers session-scoped <code>event_params</code> source/medium and uses <code>traffic_source</code> only as a documented fallback.
<code>user_pseudo_id</code>	The device/cookie id that is the de facto user key (<code>user_id</code> is almost always NULL); cross-device stitching is therefore impossible and user-grain metrics are cookie-based approximations.
Verify-then-trust	The second principle: dry-run cost + schema validation + reconciliation + adversarial Critic before any finding ships.